

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Any 4

Total Marks:40

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I (K. Deepak / 192320078) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

Question:

A Hospital Management System allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources.

Answer:

a) Actors: Patient, Doctor, Administrator, Receptionist, Payment System.

b) Use Cases: Register/Login, Book Appointment, Cancel Appointment, View Appointment, Update Medical Records, Generate Bill, Manage Resources, Manage Staff, View Reports.

c) Relationships:

- <<include>>: Book Appointment → Verify Patient; Generate Bill → Process Payment.
- <<extend>>: Cancel Appointment extends Manage Appointment.
- Generalization: Administrator and Receptionist inherit from Staff.

d) Use Case Diagram (text):

Patient → Book Appointment, View Appointment

Doctor → Update Medical Records

Administrator → Manage Resources, View Reports

Receptionist → Schedule Appointment

Payment System ↔ Process Payment

2. Class Diagram Problem

Answer:

a) Classes:

Book(bookId,title,author,status; methods: search(),updateStatus())

Member(memberId,name; borrowBook(),returnBook())

Librarian(empId,name; addBook(),removeBook(),issueBook())

IssueRecord(issueId,issueDate,dueDate; calculateFine(),closeIssue())

b) Relationships:

Association: Member-IssueRecord, Book-IssueRecord

Aggregation: Library contains Books

Inheritance: Librarian inherits Person; Member inherits Person

c) Multiplicity:

Member 1..* IssueRecords

Book 1..* IssueRecords (over time)

Librarian 1..* IssueRecords

d) Diagram (text):

Person

|-- Member

|-- Librarian

Member -- IssueRecord -- Book

3. Sequence Diagram Problem

Answer:

a) Objects: Customer, App, Order System, Payment Gateway, Restaurant.

b) Message Flow:

Customer→App: Select Food

App→Order System: Create Order

Order System→Payment Gateway: Request Payment

Payment Gateway→Order System: Payment Success

Order System→Restaurant: Send Order

Restaurant→Order System: Confirm

Order System→Customer: Order Confirmation

c) Lifelines: All objects remain active during interaction.

d) Activation: Each object activates while processing received messages.

4. Activity Diagram Problem

Answer:

a) Activities: Submit Application → Verify Documents → Pay Fee → Confirm Enrollment.

b) Decision: Documents valid? Yes→Fee Payment; No→Reject/Resubmit.

c) Parallel: Verify personal and academic documents simultaneously.

d) Flow:

Start→Submit→Fork→Verify Docs→Join→Decision→(Yes)Pay Fee→Enroll→End / (No)Reject→End

5. State Diagram Problem

Answer:

a) States: Initial, Idle, Product Selected, Payment Received, Product Dispensed, Transaction Completed, Final.

b) Events:

Select Product, Insert Money, Payment Success, Dispense Product, Finish Transaction.

c) Includes initial and final states.

d) Flow:

Initial→Idle→Product Selected→Payment Received→Product Dispensed→Transaction Completed→Final.

6. Component Diagram Problem

Answer:

a) Components: User Management, Course Management, Assessment Module, Content Delivery Module, Notification Service, Database.

b) Interfaces:

User Management↔Database

Course Management↔Content Delivery

Assessment↔Database

Notification↔User Management

c) Dependencies:

Course depends on User Management.

Assessment depends on Course.

Notification depends on Assessment and User Management.

d) Architecture:

Users→User Management→Course Management→Assessment

Course Management→Content Delivery

All modules→Database

Notification Service sends alerts to users.

7. Deployment Diagram Problem

Question:

A Smart Traffic Monitoring System uses roadside sensors to collect vehicle data...

Answer:

a) Hardware Nodes:

Roadside Sensors, Edge Gateway, Cloud Server, Database Server, Traffic Authority PC/Web Browser.

b) Software:

Sensors: Sensor Firmware

Gateway: Data Collector

Cloud: Traffic Analytics, Alert Engine, API

DB Server: Traffic Database

PC: Web Dashboard

c) Communication:

Sensors→Gateway (Wi-Fi/5G)

Gateway→Cloud (Internet)

Cloud↔Database

Traffic PC↔Cloud (HTTPS)

d) Deployment (text):

[Sensor]--[Gateway]--Internet--[Cloud Server]--[Database]

|
[Traffic Dashboard]

RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation